Module 13: How Populations Evolve - Key Points

Learning Objectives

By the end of this module, you should be able to:

1. Define microevolution using allele frequencies.
2. Compare mutation, gene flow, drift, selection, and the effects of nonrandom mating.
3. Distinguish bottleneck and founder effects.
4. Interpret directional, stabilizing, and disruptive selection.
5. Use Hardy-Weinberg as a non-evolving baseline.

Topics

- Allele frequencies as the unit of change
- Sources of variation
- Five mechanisms of microevolution
- Genetic drift and selection
- Hardy-Weinberg as a baseline

Key Terms to Know

- **Allele frequency** - Proportion of a particular allele in a population.
- **Microevolution** - Allele-frequency change across generations.
- **Mutation** - DNA sequence change that can create new alleles.
- **Gene flow** - Movement of alleles between populations.
- **Genetic drift** - Random allele-frequency change, strongest in small populations.
- **Bottleneck effect** - Drift after a sharp population reduction.
- **Founder effect** - Drift when a new population starts from few individuals.
- **Hardy-Weinberg equilibrium** - Null model for allele frequencies when no evolution occurs.

Core Contents

1. Microevolution is allele-frequency change across generations. For diploids, count allele copies and divide by twice the number of individuals.
2. Mutation can create alleles; recombination reshuffles existing alleles. Gene flow moves alleles between reproducing populations.
3. Mutation, flow, drift, and selection can change allele frequencies. Nonrandom mating alone can change genotype proportions without changing alleles.
4. Drift is chance sampling, strongest in small populations. Bottlenecks reduce existing populations; founders start new ones. Selection depends on fitness.
5. Hardy-Weinberg predicts p squared, 2pq, and q squared under explicit assumptions. A mismatch alone neither identifies a cause nor proves allele-frequency change.

Study Tips

1. Complete Lab 14 as a prediction, evidence check, and revision exercise. No internet or special equipment is needed.
2. Answer the learning questions as homework practice; show reasoning and units when a question supplies numbers.
3. Use Practice Test 03, then check its self-contained key and explain any correction in your own words.
4. Connect each answer to a mechanism and evidence, rather than a memorized definition alone.

Connected Lab

- `lab-14_how-populations-evolve.md`

Generated Visuals

- Module 13: How Populations Evolve Evidence Map
- Module 13: How Populations Evolve Reasoning Sequence
- Module 13: How Populations Evolve Retrieval and Lab Check